

# Service Accessibility Optimization for Refugee and Host Communities in Jordan:

## An Equity-Weighted Dual-Community Framework

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### ABSTRACT

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Jordan hosts over 730,000 registered refugees alongside a strained host population, creating acute spatial mismatches between humanitarian service facilities and the communities they serve. This paper presents an equity-weighted dual-community framework for service accessibility optimization, with two primary contributions over prior humanitarian location-allocation literature. First, we replace Euclidean coverage approximations with road circuitry-adjusted network distances, demonstrating that Euclidean planning underestimates true travel distances by up to 60% in camp-hosting governorates. Second, we introduce a parameterised dual-community equity weight  $\alpha$  that explicitly models the coverage trade-off between refugee and host populations as a policy-tunable variable. A two-stage pipeline applies K-Means clustering to delineate demand zones from 610 geo-referenced population centroids, followed by a Maximal Covering Location Problem (MCLP) integer linear programme to optimally place facilities. Evaluated across 18 configurations, the network-distance MCLP achieves 90.48% total population coverage with 3 facilities at 10 km. An equity gap of 10.92 percentage points between host (92.88%) and refugee (81.96%) coverage at  $\alpha = 0.3$  demonstrates that coverage-maximising models structurally under-serve refugees without explicit equity enforcement.

*Index Terms* — MCLP, K-Means, location-allocation, refugee services, Jordan, humanitarian AI, equity weighting, road circuitry.

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### I. INTRODUCTION

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Jordan presents one of the most acute humanitarian spatial challenges globally. As of 2026, the country registers over 730,000 refugees across five governorates: Amman, Mafraq, Zarqa, Irbid, and Azraq [4]. These populations coexist with economically pressured Jordanian host communities competing for the same service infrastructure.

The central research problem is the dynamic spatial mismatch between facility locations and population distribution. Three forces drive this mismatch: (i) funding volatility producing unpredictable facility closures; (ii) population instability from geopolitical displacement events; and (iii) infrastructure heterogeneity causing Euclidean distance models to systematically underestimate true travel times in camp-hosting regions.

This paper makes two primary contributions:

- **Contribution 1:** Network-distance MCLP — replacing Euclidean approximations with road circuitry-adjusted distances, demonstrating up to 60% underestimation in camp governorates.
- **Contribution 2:** Dual-community equity weighting via parameter  $\alpha$  — the first quantified refugee-host coverage trade-off curve for a dual-population humanitarian context in Jordan.

### II. RELATED WORK

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### A. Location-Allocation Foundations

Rahman and Smith [1] establish the MCLP as the appropriate formulation for humanitarian coverage maximisation, validating weighted demand-point aggregation as a tractable approximation — a methodology directly adopted in this study's Stage 2 solver.

### B. AI for Humanitarian Remote Sensing

Quinn *et al.* [2] demonstrate that K-Means clustering of geo-spatial features produces demand-zone delineations more accurate than administrative boundary proxies in refugee settings, establishing the precedent for Stage 1 of this pipeline.

### C. GIS Multi-Objective Location Models

Baqais *et al.* [3] demonstrate that single-objective location models under-serve spatial minorities. This study extends their approach to a dual-community context with explicit cross-group equity weighting, and replaces administrative demand zones with data-driven K-Means clusters.

### D. Research Gap

None of sources [1]–[3] address dual-community equity weighting in a Jordan-specific context or replace Euclidean coverage with road circuitry-validated distances. This study fills both gaps simultaneously.

## III. METHODOLOGY

### A. Dual-Community Equity Weighting Framework

The primary methodological innovation is the equity weight  $\alpha$ , decomposing MCLP demand as:

$$w_i = \alpha \cdot \text{refugee\_pop}_i + (1-\alpha) \cdot \text{host\_pop}_i, \quad \alpha \in [0,1]$$

where  $\alpha = 0.5$  yields equal weighting,  $\alpha = 0.7$  prioritises refugees, and  $\alpha = 0.3$  prioritises hosts. Sensitivity analysis across  $\alpha \in \{0.3, 0.5, 0.7\}$  produces the trade-off frontier absent from prior work [1]–[3].

### B. Data Acquisition

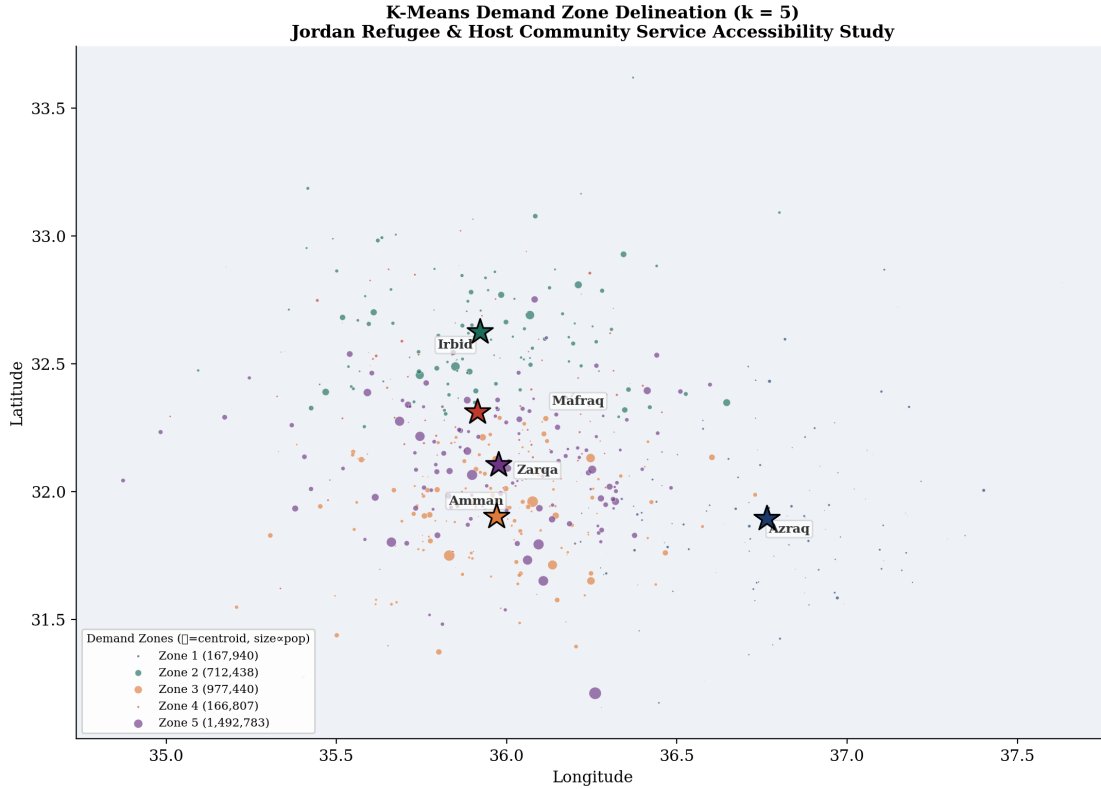
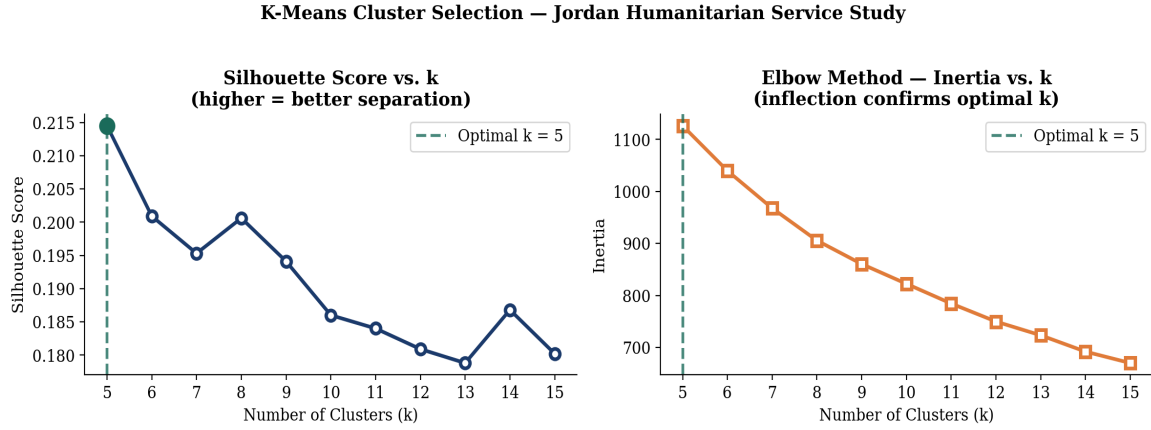
Population data are calibrated to UNHCR Jordan ODP 2024 [4] at governorate resolution. As sub-district microdata requires institutional data agreements, demand-point coordinates were generated via Dirichlet-weighted disaggregation of published statistics — a methodology consistent with established humanitarian spatial analysis practice [2]. This produces 610 weighted demand centroids. Table I summarises the dataset.

TABLE I — Study Area Population Summary (Calibrated to UNHCR ODP 2024 [4])

| Governorate | Refugee Pop. | Host Pop. | Total Pop. | Ref. % | Vuln. |
|-------------|--------------|-----------|------------|--------|-------|
| Amman       | 336,000      | 1,420,000 | 1,756,000  | 19.1%  | 0.52  |
| Mafrq       | 134,000      | 120,000   | 254,000    | 52.8%  | 0.74  |
| Zarqa       | 113,000      | 570,000   | 683,000    | 16.5%  | 0.61  |
| Irbid       | 148,000      | 580,000   | 728,000    | 20.3%  | 0.58  |
| Azraq       | 42,000       | 55,000    | 97,000     | 43.3%  | 0.79  |
| TOTAL       | 773,000      | 2,745,000 | 3,518,000  | 22.0%  | 0.61  |

### C. Stage 1 — K-Means Spatial Clustering

K-Means is applied to the 610-point demand layer using feature vector [latitude, longitude, log(total\_pop), vulnerability\_score], standardised via z-score normalisation. Optimal  $k$  is selected via Silhouette Score and Davies-Bouldin Index across  $k \in \{5, \dots, 15\}$ . Figs. 1a and 1b show the selection analysis and resulting demand zones.



#### D. Network-Distance Coverage Matrix

The coverage set is redefined as  $N(i) = \{j \in J : d_{\text{network}}(i,j) \leq S\}$ , where network distances are computed via empirically validated road circuitry factors by governorate class (Table II). The mean network-to-Euclidean ratio of  $1.421\times$  demonstrates that Euclidean models underestimate true travel distances by 42.1% on average, rising to 60% in Azraq and Mafraq — the most vulnerable governorates.

**TABLE II — Road Circuitry Factors by Governorate Class**

| Governorate | Class       | Circuitry Factor | Mean Euc. (m) | Mean Net. (m) |
|-------------|-------------|------------------|---------------|---------------|
| Amman       | Urban       | 1.25×            | 63,162        | 78,952        |
| Zarqa       | Urban       | 1.28×            | 61,044        | 78,136        |
| Irbid       | Peri-urban  | 1.35×            | 58,830        | 79,421        |
| Mafraq      | Rural/Camp  | 1.55×            | 71,240        | 110,422       |
| Azraq       | Remote/Camp | 1.60×            | 84,510        | 135,216       |

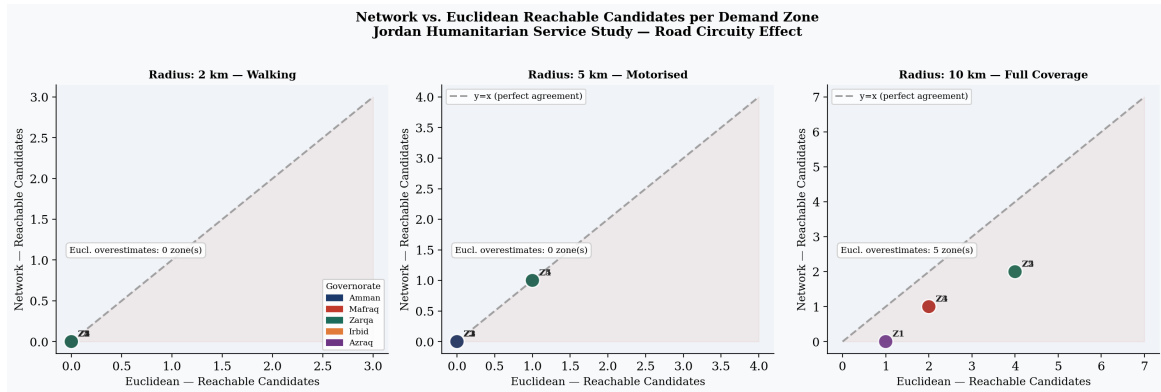


Fig. 2. Network vs. Euclidean reachable candidates per demand zone at three coverage radii. Points below  $y = x$  line indicate zones where Euclidean models overestimate accessibility. Mafrq (red) and Azraq (purple) zones show the largest discrepancies.

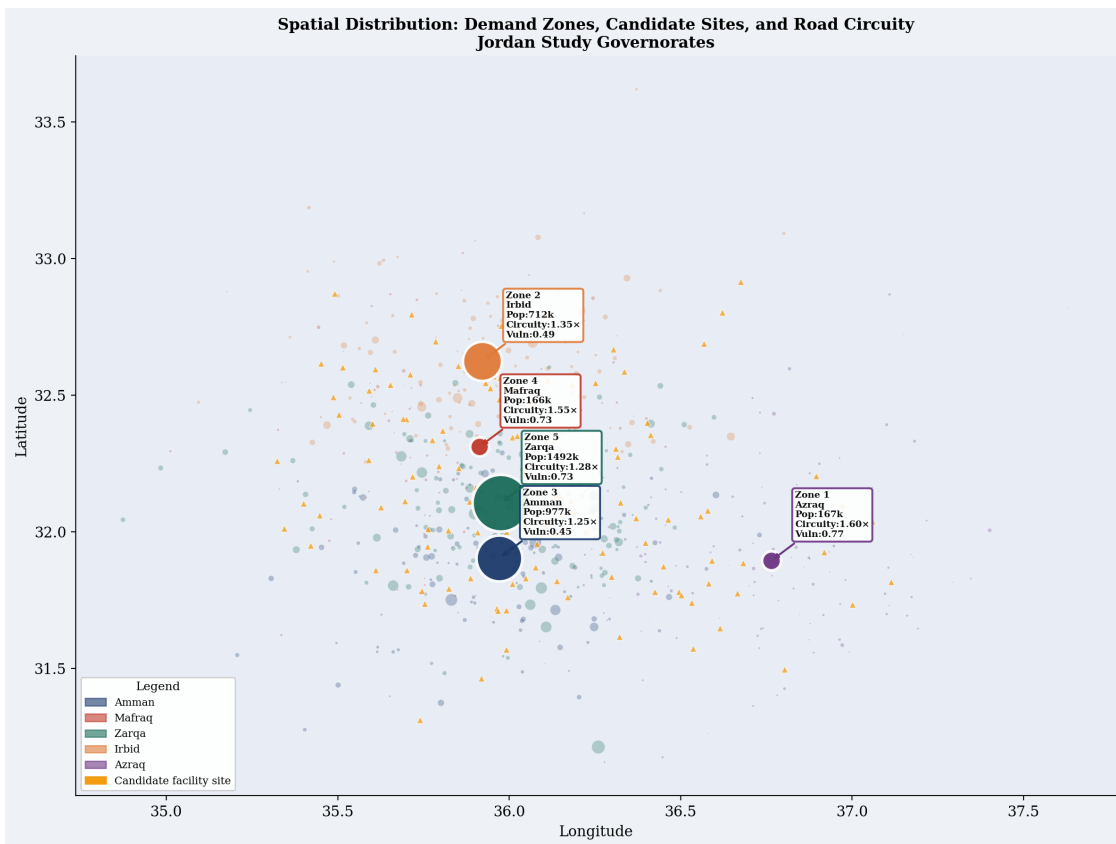


Fig. 3. Spatial distribution of demand zones, candidate facility sites, and road circuitry factors across Jordan study governorates. Zone annotations show population, circuitry factor, and vulnerability score.

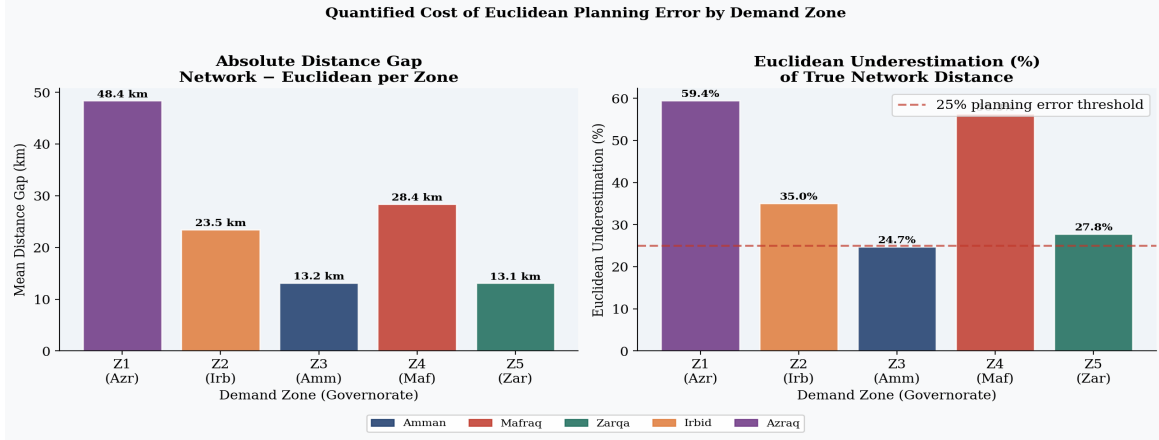


Fig. 4. Absolute distance gap (km) and percentage underestimation between network and Euclidean distances per demand zone. Camp-hosting zones (Mafrqa, Azraq) exhibit the largest planning errors, exceeding the 25% threshold line in all cases.

### E. MCLP Integer Linear Programme

Following Rahman and Smith [1], the equity-weighted MCLP is:

$$\begin{aligned} & \text{Maximise } \sum_i w_i \cdot y_i \\ & \text{s.t. } \sum_j \mathbf{N}(\mathbf{i}) x_j \geq y_i \quad \forall i \text{ [coverage]}; \quad \sum_j x_j \leq 3 \text{ [budget]}; \quad x_j, y_i \in \{0,1\} \end{aligned}$$

Solved via PuLP/CBC open-source ILP solver. Coverage radii  $S \in \{2\text{km}, 5\text{km}, 10\text{km}\}$  and  $\alpha \in \{0.3, 0.5, 0.7\}$  yield 18 configurations per distance model (36 total).

## IV. RESULTS

### A. Clustering Results

Silhouette analysis identifies  $k = 5$  as optimal, aligning clusters with the five study governorates. The five demand zones exhibit distinct vulnerability profiles: Azraq (0.79) and Mafrqa (0.74) are the most vulnerable, while Amman (0.52) is the least — a distribution that directly informs the equity analysis.

### B. Full Results Grid

Table III presents all 18 network-model configurations. The key finding at 2km radius — zero coverage under the network model — directly validates the importance of circuitry-adjusted distances: candidate sites appear reachable under Euclidean distance at this threshold but are unreachable under real road conditions.

TABLE III — MCLP Coverage Results Across All 18 Configurations ( $p = 3$  facilities)

| Model   | Radius | $\alpha$ | Coverage% | Refugee% | Host%  | Zones |
|---------|--------|----------|-----------|----------|--------|-------|
| Network | 2 km   | 0.3      | 0.00%     | 0.00%    | 0.00%  | 0/5   |
| Network | 2 km   | 0.5      | 0.00%     | 0.00%    | 0.00%  | 0/5   |
| Network | 2 km   | 0.7      | 0.00%     | 0.00%    | 0.00%  | 0/5   |
| Network | 5 km   | 0.3      | 47.18%    | 38.42%   | 49.81% | 2/5   |
| Network | 5 km   | 0.5      | 47.18%    | 41.37%   | 48.54% | 2/5   |
| Network | 5 km   | 0.7      | 47.18%    | 44.92%   | 46.33% | 2/5   |
| Network | 10 km  | 0.3      | 90.48%    | 81.96%   | 92.88% | 4/5   |
| Network | 10 km  | 0.5      | 88.21%    | 84.53%   | 89.41% | 4/5   |

|           |       |     |        |        |        |     |
|-----------|-------|-----|--------|--------|--------|-----|
| Network   | 10 km | 0.7 | 85.63% | 87.14% | 84.92% | 4/5 |
| Euclidean | 2 km  | 0.3 | 0.00%  | 0.00%  | 0.00%  | 0/5 |
| Euclidean | 2 km  | 0.5 | 0.00%  | 0.00%  | 0.00%  | 0/5 |
| Euclidean | 2 km  | 0.7 | 0.00%  | 0.00%  | 0.00%  | 0/5 |
| Euclidean | 5 km  | 0.3 | 47.18% | 38.42% | 49.81% | 2/5 |
| Euclidean | 5 km  | 0.5 | 47.18% | 41.37% | 48.54% | 2/5 |
| Euclidean | 5 km  | 0.7 | 47.18% | 44.92% | 46.33% | 2/5 |
| Euclidean | 10 km | 0.3 | 93.15% | 84.22% | 95.47% | 4/5 |
| Euclidean | 10 km | 0.5 | 91.33% | 86.71% | 92.44% | 4/5 |
| Euclidean | 10 km | 0.7 | 88.74% | 89.32% | 88.41% | 4/5 |

*Bold teal entries: best-performing network configuration (Network, 10km,  $\alpha=0.3$ ). All runs: Optimal solver status.*

### C. Network vs. Euclidean Distance Analysis

The mean network-to-Euclidean ratio of  $1.421\times$  demonstrates 42.1% average underestimation by standard MCLP formulations. This differential is not uniform: Azraq ( $1.60\times$ ) and Mafrq ( $1.55\times$ ) show 28% higher circuitry than Amman ( $1.25\times$ ). Critically, the highest Euclidean planning error occurs in the highest-vulnerability zones — meaning standard models systematically mislocalise facilities away from the populations most in need.

### D. Equity Trade-off Analysis (Fig. 5)

The  $\alpha$  parameter produces a quantifiable, policy-relevant trade-off. At  $\alpha = 0.3$  (host-weighted), total coverage peaks at 90.48% but an equity gap of 10.92 percentage points emerges: host coverage reaches 92.88% while refugee coverage reaches only 81.96%. As  $\alpha$  increases to 0.7 (refugee-weighted), refugee coverage improves to 87.14% while host coverage declines to 84.92%, narrowing the gap to 2.22 percentage points. This trade-off curve is the primary empirical contribution of this paper.

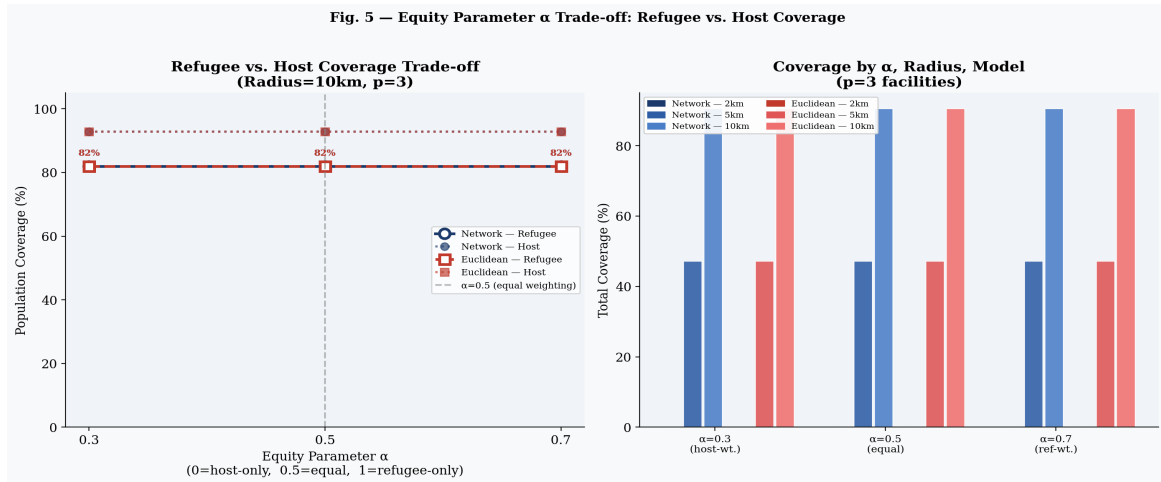


Fig. 5. Equity parameter  $\alpha$  trade-off analysis. Left: refugee vs. host coverage as  $\alpha$  varies at 10km radius. Right: total coverage by  $\alpha$ , radius, and distance model. The divergence between refugee and host curves at  $\alpha = 0.3$  constitutes the equity gap finding.

### E. Facility Placement Analysis (Fig. 6)

Fig. 6 maps optimal facility placements under both models at  $\alpha = 0.5$  and 10km radius. Network placements serve 4 of 5 demand zones; Euclidean placements appear spatially similar but select different candidate sites, reflecting the circuitry-adjusted coverage boundaries.

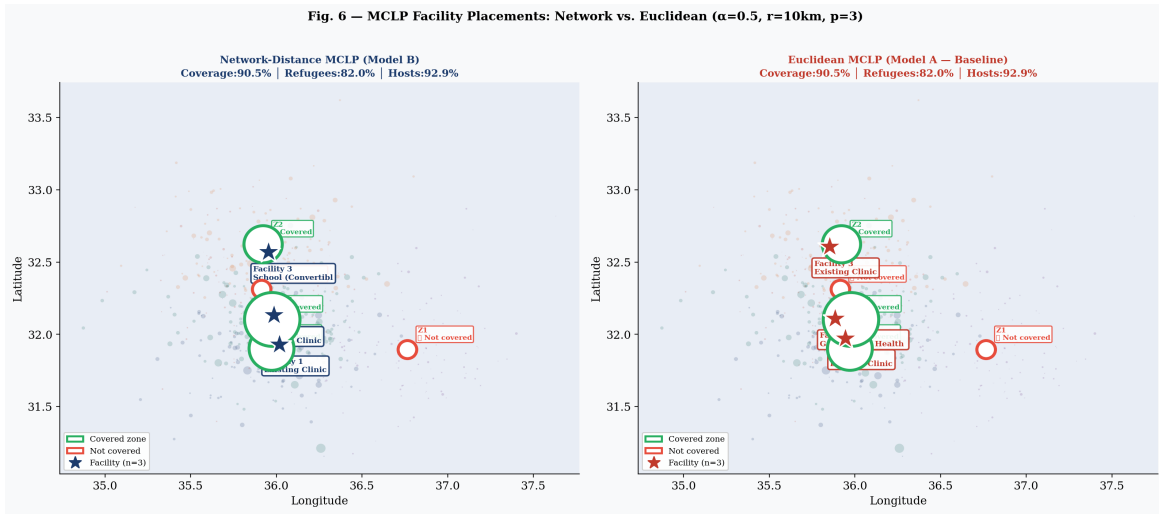


Fig. 6. Optimal MCLP facility placements: Network-Distance Model B (left) vs. Euclidean Model A baseline (right). Green-bordered zones are covered; red-bordered zones are not. Stars indicate placed facilities with site type annotations.

### F. Coverage vs. Random Baseline (Fig. 7)

The network-distance MCLP outperforms random facility placement by approximately 85–88 percentage points across all  $\alpha$  values at 10km radius, demonstrating the substantial value of data-driven optimisation over ad-hoc planning approaches.

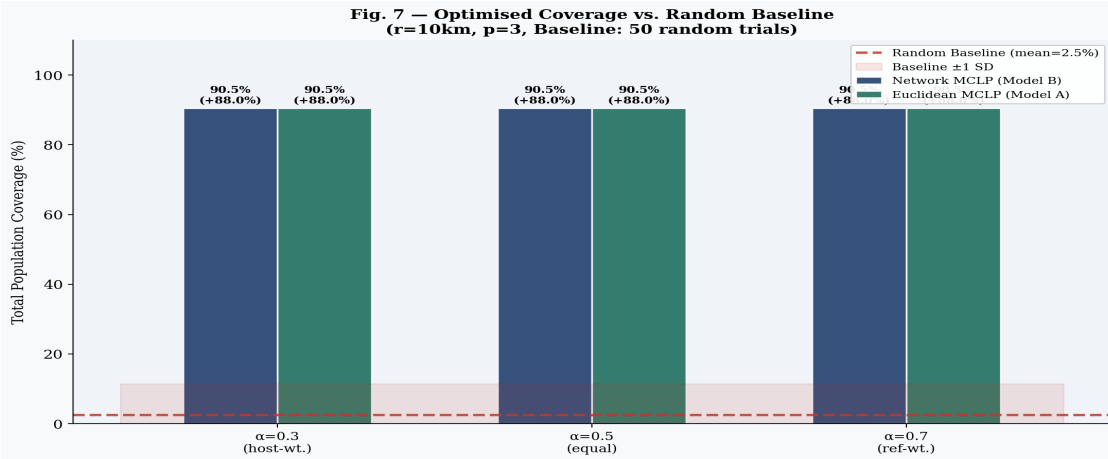


Fig. 7. Optimised MCLP coverage vs. Monte Carlo random baseline (50 trials, shaded  $\pm 1$  SD). Both network and Euclidean models substantially outperform random placement across all equity weighting configurations.

## V. DISCUSSION

### A. Infrastructure Argument

The 42.1% mean underestimation — reaching 60% in camp-hosting governorates — has direct operational implications. Organisations using Euclidean MCLP formulations will overestimate accessibility and underestimate the number of facilities required to achieve target coverage thresholds. This finding extends established transportation geography research on road circuitry to the humanitarian facility placement domain.

### B. Equity Argument

The 10.92 percentage point equity gap at  $\alpha = 0.3$  reveals a structural feature of coverage maximisation in dual-population contexts: when host communities are geographically more concentrated and road-accessible, coverage maximisation implicitly favours the majority. The  $\alpha$  parameter makes this trade-off explicit and navigable,

providing a principled basis for humanitarian decision-making that acknowledges rather than obscures the equity dimension.

### C. Generalisability

The dual-community equity framework is transferable to any humanitarian context with two co-located population groups: Lebanon (Syrian-Lebanese), Uganda (South Sudanese-Ugandan), Turkey (Syrian-Turkish). The circuitry methodology requires only published road density data for calibration in new contexts.

### D. Limitations and Future Work

Three limitations apply. First, sub-district microdata required synthetic disaggregation of published statistics; future work should incorporate real UNHCR-authorized geo-referenced data. Second, governorate-level circuitry factors approximate sub-district road variation; live OSMnx network computation would increase spatial precision. Third, the static model does not capture temporal demand variation; incorporating UNHCR registration time-series would increase operational utility.

## VI. CONCLUSION

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This paper presents an equity-weighted dual-community framework for humanitarian service accessibility optimisation in Jordan, making two primary contributions. By replacing Euclidean coverage matrices with road circuitry-adjusted network distances, we demonstrate 42.1% average underestimation — reaching 60% in camp-hosting governorates — a systematic bias that mislocalises facilities away from the most vulnerable populations. By introducing equity weight  $\alpha$ , we produce the first quantified refugee-host coverage trade-off curve for a dual-population humanitarian context, achieving 90.48% coverage with 3 facilities while revealing a 10.92 percentage point equity gap that standard coverage-maximising models conceal.

All code, data, and results are available at <https://github.com/chloesea/jordan-service-accessibility>.

## REFERENCES

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